The spherical average process and quantum cones in arbitrary odd dimension

Abstract

For $d > 2$ odd and $\gamma \in (0, \sqrt{2d})$, we identify the spherical average process of the whole-space log-correlated Gaussian field on $\mathbb{R}^d$ as an integral transform of a two-sided Brownian motion. We then construct the d-dimensional β-quantum cone for each $\beta \in (-\infty, Q)$, where $Q = d/\gamma + \gamma/2$. More precisely, if $(S_t)_{t\in\mathbb{R}}$ is the spherical average process and

$$\sigma_b = \inf\{t \in \mathbb{R} : S_t - (Q-\beta)t = -b\}, \qquad S_{b,s} = S_{s+\sigma_b} - S_{\sigma_b},$$

then, for each fixed $T \in \mathbb{R}$, the process $(S_{b,s})_{s\geq T}$ converges in total variation distance as $b \to \infty$. Combining the limiting radial process with the projection of the whole-space field having zero average on each sphere gives the odd-dimensional quantum cone. We prove that this field is the total variation local limit obtained by appropriately recentering and rescaling a whole-space field with a β-log singularity. The main difference from the even-dimensional argument is that the conditional law of the future is described by a function of the entire past rather than by a finite vector of derivatives. We obtain the required comparison estimate by inverting the corresponding integral kernel on the half-line.

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1 Introduction

Let h be the whole-space log-correlated Gaussian field (LGF) on $\mathbb{R}^d$, viewed modulo an additive constant. Formally, the covariance kernel of h is

$$\mathbb{E}[h(x)h(y)] = -\log|x-y|. \tag{1.1}$$

The field can be defined rigorously as a Gaussian random distribution on test functions of integral zero; see Section 3. We choose the additive constant so that the average of h over the unit sphere centred at the origin is zero. If $h_r(0)$ denotes the average over $\partial B(0,r)$, we set

$$S_t \stackrel{\text{def}}{=} h_{\mathrm{e}^{-t}}(0), \qquad t \in \mathbb{R}. \tag{1.2}$$

In dimension two, $(S_t)_{t\in\mathbb{R}}$ is a standard two-sided Brownian motion. This observation is one of the basic inputs in the construction of the two-dimensional quantum cone in [4].

For even dimensions $d > 2$, Bertacco and Gwynne [2] showed that the spherical average process can be identified with the integral of a stationary Gaussian Markov process of order $(d-2)/2$. They used this finite-dimensional Markov representation to construct the d-dimensional quantum cone and to prove that it arises as a local limit of a whole-space LGF with a logarithmic singularity. The finite-dimensional representation is special to even dimensions. Indeed, in that case the power spectrum of the generalized derivative of S is rational. In odd dimensions, the corresponding power spectrum contains an additional hyperbolic factor and does not admit the same finite-dimensional realization.

The first result of this paper is an explicit representation which is valid in every odd dimension. Suppose that

$$d = 2m+1, \qquad m \in \mathbb{N}, \qquad \rho \stackrel{\text{def}}{=} \frac{d-2}{2} = m - \frac{1}{2}. \tag{1.3}$$

If B is a standard two-sided Brownian motion, then

$$S_t \stackrel{\text{law}}{=} \int_0^\infty q_d(u)(B_{t-u} - B_{-u})\,\mathrm{d}u, \qquad q_d(u) = (d-2)\mathrm{e}^{-2u}(1-\mathrm{e}^{-2u})^{(d-4)/2}\mathbf{1}_{\{u>0\}}. \tag{1.4}$$

The proof follows from an exact spectral factorisation of the covariance of the spherical average process. Formula (1.4) is the odd-dimensional counterpart of the representation in [2, Theorem A].

Fix $\gamma \in (0, \sqrt{2d})$ and set

$$Q \stackrel{\text{def}}{=} \frac{d}{\gamma} + \frac{\gamma}{2}. \tag{1.5}$$

For $\beta < Q$, the β-quantum cone is obtained by recentering the process $S_t - (Q-\beta)t$ when it first reaches a large negative level. In dimension two, convergence of the recentered process is a consequence of the strong Markov property of Brownian motion. In even dimensions, the proof in [2] uses the Markov property of the vector consisting of the first $(d-2)/2$ derivatives of S.

For odd d, the future of S conditional on the past is not determined by a finite vector. Nevertheless, formula (1.4) gives an exact description of the conditional law. If τ is a stopping time for the Brownian motion in (1.4), then

$$S_{\tau+t} - S_\tau = \int_0^t K_d(t-s)\,\mathrm{d}W_s + M_\tau(t) - M_\tau(0), \qquad K_d(t) = (1-\mathrm{e}^{-2t})^\rho, \tag{1.6}$$

where $W_t = B_{\tau+t} - B_\tau$ is independent of the past and M_τ is a random function determined by the Brownian path before τ. Thus, the role played by the finite derivative vector in even dimensions is played here by the function M_τ.

The main analytic input is a comparison estimate for the conditional laws in (1.6). The Laplace transform of $q_d = K_d'$ is

$$\mathcal{Q}_d(z) = \Gamma(d/2)\frac{\Gamma(1+z/2)}{\Gamma(d/2+z/2)}, \qquad \Re z > 0. \tag{1.7}$$

Since $\mathcal{Q}_d(z)$ is of order $(1+z)^{-\rho}$, division by $\mathcal{Q}_d$ maps the appropriate half-line Sobolev space into $L^2(0,\infty)$. This allows us to change the driving Brownian motion by a Cameron–Martin shift which exactly cancels the difference between two possible functions M_τ. The resulting total variation estimate tends to zero when the observation time tends to infinity.

We then compare the hitting times of $S_t - (Q-\beta)t$ with the corresponding hitting times for the Brownian radial process used in the two-dimensional cone construction. This gives tightness of the function M_τ at the recentering times. The comparison estimate therefore implies that the law of the recentered process becomes asymptotically independent of the past. A coupling argument gives convergence in total variation on each interval $[T,\infty)$.

Finally, the homogeneous Sobolev space associated with the whole-space LGF decomposes orthogonally into its radially symmetric subspace and the subspace of functions whose average on each centred sphere is zero. The recentering time depends only on the radial projection. The second projection is unchanged in law by dilations and remains independent of the radial part. Hence the convergence of the radial process immediately gives convergence of the fields on each fixed ball.

The rest of the paper is organised as follows. We state the main results in Section 2. Section 3 recalls the whole-space LGF, spherical averages, and the radial–spherical decomposition. In Section 4, we prove the Brownian representation (1.4). Section 5 gives the conditional law of the future and proves the comparison estimate. In Section 6, we prove convergence of the recentered process, define the odd-dimensional quantum cone, and establish the local limit theorem. Some analytic and measure-theoretic facts used in the proof are collected in the appendices.

2 Main results

Throughout the paper, $d = 2m+1 > 2$ is odd, h is a whole-space LGF on $\mathbb{R}^d$ with $h_1(0) = 0$, and $(S_t)_{t\in\mathbb{R}}$ is defined as in (1.2). All spaces of continuous functions are equipped with the topology of uniform convergence on compact sets and the corresponding Borel σ-field. We use the convention

$$d_{\mathrm{TV}}(\mu,\nu) \stackrel{\mathrm{def}}{=} \sup_A |\mu(A) - \nu(A)|. \tag{2.1}$$

2.1 Identification of the spherical average in arbitrary odd dimension

For $t > 0$, define

$$K_d(t) \stackrel{\mathrm{def}}{=} (1-\mathrm{e}^{-2t})^\rho, \qquad q_d(t) \stackrel{\mathrm{def}}{=} K_d'(t) = (d-2)\mathrm{e}^{-2t}(1-\mathrm{e}^{-2t})^{(d-4)/2}, \tag{2.2}$$

and extend q_d by zero to $(-\infty, 0]$.

Theorem A (Identification of the spherical average)**.** *Let $B = (B_t)_{t\in\mathbb{R}}$ be a standard two-sided Brownian motion with $B_0 = 0$. Then*

$$(S_t)_{t\in\mathbb{R}} \overset{\text{law}}{=} \left(\int_0^\infty q_d(u)(B_{t-u} - B_{-u})\,\mathrm{d}u\right)_{t\in\mathbb{R}} \tag{2.3}$$

as random elements of $C_{\mathrm{loc}}(\mathbb{R})$. The integral in (2.3) is an ordinary pathwise Lebesgue integral.

The one-sided Fourier transform of q_d is

$$Q_d(\xi) \overset{\text{def}}{=} \int_0^\infty q_d(t)\mathrm{e}^{-\mathrm{i}\xi t}\,\mathrm{d}t = \Gamma(d/2)\frac{\Gamma(1+\mathrm{i}\xi/2)}{\Gamma(d/2+\mathrm{i}\xi/2)}. \tag{2.4}$$

In particular, for $d = 2m+1$,

$$|Q_d(\xi)|^2 = 2^{d-2}\Gamma(d/2)^2\frac{\xi\coth(\pi\xi/2)}{\prod_{j=0}^{m-1}\left(\xi^2+(2j+1)^2\right)}, \tag{2.5}$$

with the value at $\xi = 0$ defined by continuity. The factor $\coth(\pi\xi/2)$ is absent in even dimensions and is the reason why the finite-dimensional argument of [2] does not apply directly.

2.2 Quantum cones in arbitrary odd dimension

Fix $\gamma \in (0, \sqrt{2d})$, let Q be as in (1.5), and fix $\beta \in (-\infty, Q)$. We write

$$a \overset{\text{def}}{=} Q - \beta > 0. \tag{2.6}$$

For $b > 0$, define

$$\sigma_b \overset{\text{def}}{=} \inf\{t \in \mathbb{R} : S_t - at = -b\} \tag{2.7}$$

and

$$S_{b,s} \overset{\text{def}}{=} S_{s+\sigma_b} - S_{\sigma_b}, \qquad s \in \mathbb{R}. \tag{2.8}$$

The sublinear growth estimate proved in Proposition 4.4 implies that σ_b is finite almost surely.

Theorem B (Construction of the radial part of the quantum cone)**.** *For each fixed $T \in \mathbb{R}$, the process $(S_{b,s})_{s\geq T}$ converges in total variation distance, as $b \to \infty$, to a continuous process $(S_{\infty,s})_{s\geq T}$. These limiting processes are compatible as T varies and hence determine a continuous process $(S_{\infty,s})_{s\in\mathbb{R}}$ with $S_{\infty,0} = 0$.*

Let $\mathcal{H}_d$ be the homogeneous Sobolev space of order $d/2$ associated with the whole-space LGF. As recalled in Lemma 3.1,

$$\mathcal{H}_d = \mathcal{H}_{d,\mathrm{rad}} \oplus \mathcal{H}_{d,\mathrm{sph}}, \tag{2.9}$$

where $\mathcal{H}_{d,\mathrm{rad}}$ is the closure of radially symmetric Schwartz functions modulo constants and $\mathcal{H}_{d,\mathrm{sph}}$ is the closure of Schwartz functions having zero average on each sphere centred at the origin.

Definition 2.1 (Odd-dimensional quantum cone)**.** Let $h^\star_{\mathrm{sph}}$ have the law of the projection of the whole-space LGF onto $\mathcal{H}_{d,\mathrm{sph}}$, independently of S_∞. The d-dimensional β-quantum cone field is

$$h^\star \overset{\text{def}}{=} h^\star_{\mathrm{rad}} + h^\star_{\mathrm{sph}}, \tag{2.10}$$

where $h^\star_{\mathrm{rad}}$ is constant equal to $S_{\infty,s} + \beta s$ on the sphere $\partial B(0, \mathrm{e}^{-s})$, for each $s \in \mathbb{R}$. The additive constant is fixed by the condition $h^\star_1(0) = 0$.

We next state the local limit theorem. For a deterministic $z \in \mathbb{R}^d$, normalize the field around z by subtracting $h_1(z)$. For $t \in \mathbb{R}$ and $b > 0$, let

$$h^{t,b}(x) \stackrel{\text{def}}{=} h(z + \mathrm{e}^{-t}x) - h_1(z) - \beta \log |\mathrm{e}^{-t}x| - Qt + b. \tag{2.11}$$

Let

$$\sigma_b \stackrel{\text{def}}{=} \inf\{t \in \mathbb{R} : h_1^{t,b}(0) = 0\}. \tag{2.12}$$

This agrees with (2.7) for the spherical average process centred at z.

Theorem C (Quantum cone as a local limit)**.** *For each $R > 0$, the restriction $h^{\sigma_b,b}|_{B(0,R)}$ converges in total variation distance to $h^\star|_{B(0,R)}$ as $b \to \infty$. The laws are viewed as probability measures on the space of distributions on $B(0, R)$.*

2.3 Outline and comparison with the even-dimensional case

The proof of Theorem A is given in Section 4. We start from the covariance formula for the spherical averages of the whole-space LGF and compute the Fourier transform of its mixed derivative. The resulting spectral density is the squared modulus of the Fourier transform of q_d, which proves (2.3). This calculation is valid for every integer $d > 2$; the restriction to odd dimension enters only when we analyse the regularity and inverse of q_d.

The proof of Theorem B is carried out in Sections 5 and 6. If $d > 2$ is even, the vector of derivatives of S up to order $(d-2)/2$ is Markov and the convergence argument of [2] reduces to a finite-dimensional mixing estimate. If d is odd, the conditional law after a stopping time is given by (1.6); the dependence on the past is represented by a function M_τ on $[0, \infty)$. We place this function in a weighted Sobolev space $\mathcal{E}_{\lambda,m}$ and prove that the conditional laws corresponding to two elements $M, N \in \mathcal{E}_{\lambda,m}$ satisfy

$$d_{\mathrm{TV}}(P_u^M, P_u^N) \le C_{d,\lambda} \left(\frac{|M(0) - N(0)|}{\sqrt{u}} + \mathrm{e}^{-\lambda u} \|M - N\|_{\mathcal{E}_{\lambda,m}} \right). \tag{2.13}$$

The estimate follows from a deterministic change of the driving Brownian motion. The required change is obtained by solving an integral equation with kernel K_d. Differentiating the equation reduces it to convolution by q_d, and the quotient of Gamma functions in (1.7) gives a bounded Hardy-space multiplier after multiplication by $(1 + z)^{-\rho}$.

To use (2.13), we must control the function M_{σ_b} at the random times (2.7). We compare S with the image under (2.3) of the two-sided Brownian path used in the two-dimensional cone construction. The latter path is invariant in law under recentering at its level-b hitting time. The difference between the two corresponding hitting times is tight, and the difference between their past contributions tends to zero. This yields tightness of M_{σ_b} in $\mathcal{E}_{\lambda,m}$.

The comparison estimate then shows that, a long deterministic time after the level-b hitting time, the recentered process is close in total variation to an ordinary copy of S. A second hitting-time argument compares the level-$(b + c)$ and level-c recentered laws, which proves that these laws are Cauchy in total variation. The field local limit in Theorem C follows from the orthogonal decomposition (2.9) in the same manner as in [2, Theorem D].

3 Background and preliminaries

3.1 The whole-space log-correlated Gaussian field

Let

$$\mathcal{S}_0(\mathbb{R}^d) \stackrel{\text{def}}{=} \left\{ \varphi \in \mathcal{S}(\mathbb{R}^d) : \int_{\mathbb{R}^d} \varphi(x)\,\mathrm{d}x = 0 \right\}.$$

The whole-space LGF is the centred Gaussian random distribution modulo constants characterised by

$$\mathbb{E}[(h,\varphi)(h,\psi)] = -\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \log|x-y|\varphi(x)\psi(y)\,\mathrm{d}x\,\mathrm{d}y, \qquad \varphi,\psi \in \mathcal{S}_0(\mathbb{R}^d). \tag{3.1}$$

Equivalently, it is the standard Gaussian associated with the homogeneous Sobolev space $\mathcal{H}_d$ of order $d/2$. We use the normalisation of [2]; the precise multiplicative constant in the Sobolev inner product is chosen so that (3.1) holds.

For a conformal automorphism ϕ of $\mathbb{R}^d$ obtained as a composition of translations, dilations, and rotations, the coordinate change convention for the higher-dimensional LQG field takes the form

$$h \longmapsto h \circ \phi + Q \log|\phi'|, \qquad Q = \frac{d}{\gamma} + \frac{\gamma}{2}. \tag{3.2}$$

Only the dilation part of (3.2) is used in this paper.

The field h is invariant under translations and rotations. Under the dilation $x \mapsto \mathrm{e}^{-t}x$, the field $h(\mathrm{e}^{-t}\cdot)$ has the same law as h modulo an additive constant. This is immediate from (3.1), since

$$-\log|\mathrm{e}^{-t}x - \mathrm{e}^{-t}y| = -\log|x-y| + t.$$

The last term disappears against test functions of integral zero.

3.2 Spherical averages and the radial–spherical decomposition

Let σ_{d-1} be probability surface measure on $\mathbb{S}^{d-1}$. For $r > 0$, the spherical average $h_r(0)$ is the Gaussian random variable obtained by pairing h with the uniform probability measure on $\partial B(0,r)$, interpreted by approximation. Differences $h_r(0) - h_s(0)$ are well-defined without fixing the additive constant. We choose the additive constant so that $h_1(0) = 0$ and define S_t as in (1.2).

If U, V are independent and uniformly distributed on $\mathbb{S}^{d-1}$, the angle Θ between them has density

$$\frac{\Gamma(d/2)}{\sqrt{\pi}\,\Gamma((d-1)/2)}(\sin\theta)^{d-2}\,\mathrm{d}\theta, \qquad 0 < \theta < \pi. \tag{3.3}$$

Consequently, the covariance of two spherical averages is obtained by integrating the logarithmic kernel against (3.3). More precisely, up to terms depending on only one of the two radii, it is

$$-c_d \int_0^\pi \log\bigl(r_1^2 + r_2^2 - 2r_1r_2\cos\theta\bigr)(\sin\theta)^{d-2}\,\mathrm{d}\theta, \qquad c_d \stackrel{\text{def}}{=} \frac{\Gamma(d/2)}{2\sqrt{\pi}\,\Gamma((d-1)/2)}. \tag{3.4}$$

This is the covariance formula used in [2, Section 3.2].

We next recall the decomposition of the Gaussian Hilbert space into radial and non-radial parts. Let $\mathcal{H}_{d,\mathrm{rad}}$ be the closure in $\mathcal{H}_d$ of radially symmetric Schwartz functions, viewed modulo constants. Let $\mathcal{H}_{d,\mathrm{sph}}$ be the closure of Schwartz functions whose integral over every centred sphere is zero.

Lemma 3.1 (Radial–spherical decomposition)**.** *For every $d \geq 2$,*

$$\mathcal{H}_d = \mathcal{H}_{d,\mathrm{rad}} \oplus \mathcal{H}_{d,\mathrm{sph}}. \tag{3.5}$$

Consequently,

$$h = h_{\mathrm{rad}} + h_{\mathrm{sph}}, \tag{3.6}$$

where the two Gaussian projections are independent. The process $(S_t)_{t\in\mathbb{R}}$ is measurable with respect to h_{rad}, while h_{sph} has zero average on every centred sphere.

Proof. This is [2, Lemma 3.5]. Orthogonality follows by averaging one factor over rotations and using the rotation invariance of the Sobolev inner product. The resulting averaging operator is the orthogonal projection onto the radial subspace. The Gaussian decomposition (3.6) is the corresponding orthogonal decomposition of the isonormal Gaussian process. □

Dilations preserve both subspaces in (3.5). In particular, for each deterministic $t \in \mathbb{R}$,

$$h_{\mathrm{sph}}(\mathrm{e}^{-t}\cdot) \overset{\mathrm{law}}{=} h_{\mathrm{sph}}. \tag{3.7}$$

Since a random dilation used below is measurable with respect to h_{rad}, independence implies that (3.7) remains valid after conditioning on the dilation factor.

3.3 A Brownian path invariant under recentering

We recall the one-dimensional path which appears in the two-dimensional quantum cone construction. Fix $a > 0$. Let $\widetilde{B} = (\widetilde{B}_t)_{t\in\mathbb{R}}$ be defined as follows. The positive-time process $(\widetilde{B}_t)_{t\geq 0}$ is a standard Brownian motion started from zero. Independently, the process

$$Y_s \overset{\mathrm{def}}{=} \widetilde{B}_{-s} + as, \qquad s \geq 0, \tag{3.8}$$

has the law of Brownian motion with drift a, started from zero and conditioned to stay positive. Thus

$$\widetilde{B}_t - at > 0, \qquad t < 0. \tag{3.9}$$

For $b > 0$, define

$$\tau_b \overset{\mathrm{def}}{=} \inf\{t \in \mathbb{R} : \widetilde{B}_t - at = -b\}. \tag{3.10}$$

Then $\tau_b \geq 0$ and

$$(\widetilde{B}_{\tau_b+t} - \widetilde{B}_{\tau_b})_{t\in\mathbb{R}} \overset{\mathrm{law}}{=} (\widetilde{B}_t)_{t\in\mathbb{R}}. \tag{3.11}$$

This is the radial recentering identity from [4, Proposition 4.13]. We will use only (3.9), (3.11), and the almost sure sublinear growth

$$\frac{\widetilde{B}_t}{|t|} \longrightarrow 0 \qquad (|t| \to \infty). \tag{3.12}$$

3.4 Notation for Sobolev and Hardy spaces

For $s \geq 0$, we use the Bessel-potential norm

$$\|f\|_{H^s(\mathbb{R})}^2 \stackrel{\text{def}}{=} \int_{\mathbb{R}} (1+\xi^2)^s |\widehat{f}(\xi)|^2 \, \mathrm{d}\xi, \tag{3.13}$$

where

$$\widehat{f}(\xi) \stackrel{\text{def}}{=} (2\pi)^{-1/2} \int_{\mathbb{R}} f(t) \mathrm{e}^{-\mathrm{i}\xi t} \, \mathrm{d}t. \tag{3.14}$$

We write $H^2(\mathbb{C}_+)$ and $H^\infty(\mathbb{C}_+)$ for the Hardy spaces on the right half-plane $\mathbb{C}_+ = \{z : \Re z > 0\}$. The Paley–Wiener theorem identifies $L^2(0,\infty)$ with $H^2(\mathbb{C}_+)$ by the Laplace transform. We use the corresponding Sobolev form in Proposition 5.7 below.

4 Identification of the spherical average in arbitrary odd dimension

4.1 Fourier transform of the covariance

Put

$$\psi_\theta(u) \stackrel{\text{def}}{=} \frac{1 - \cos\theta \cosh u}{(\cosh u - \cos\theta)^2}, \qquad u \in \mathbb{R}, \quad 0 < \theta < \pi. \tag{4.1}$$

Since

$$\frac{\mathrm{d}^2}{\mathrm{d}u^2} \log\bigl(2(\cosh u - \cos\theta)\bigr) = \psi_\theta(u),$$

the mixed distributional derivative of the covariance of S is the stationary distribution

$$\mathcal{K}_d^{(1)}(u) = c_d \int_0^\pi \psi_\theta(u) (\sin\theta)^{d-2} \, \mathrm{d}\theta. \tag{4.2}$$

There is no delta mass at $u = 0$. Indeed, the only possible singularity comes from θ near zero, and for $|u| \leq 1$,

$$\frac{|\sinh u|}{\cosh u - \cos\theta} \leq C \frac{|u|}{u^2 + \theta^2}.$$

After multiplication by $(\sin\theta)^{d-2}$ and integration in θ, the first derivative of the angularly averaged covariance is continuous at zero and vanishes there.

Lemma 4.1 (Fourier transform of the covariance derivative)**.** *For every integer $d > 2$ and $\xi \in \mathbb{R}$,*

$$\widehat{\mathcal{K}_d^{(1)}}(\xi) = \frac{1}{\sqrt{2\pi}} \Gamma(d/2)^2 \frac{|\Gamma(1 + \mathrm{i}\xi/2)|^2}{|\Gamma(d/2 + \mathrm{i}\xi/2)|^2}. \tag{4.3}$$

Proof. We first use the contour integral

$$\int_{\mathbb{R}} \frac{\mathrm{e}^{-\mathrm{i}\xi u}}{\cosh u - \cos\theta} \, \mathrm{d}u = \frac{2\pi \sinh((\pi - \theta)\xi)}{\sin\theta \sinh(\pi\xi)}, \qquad 0 < \theta < \pi, \tag{4.4}$$

where the value at $\xi = 0$ is obtained by continuity. To prove (4.4), integrate $z \mapsto \mathrm{e}^{-\mathrm{i}\xi z}/(\cosh z - \cos\theta)$ around a rectangle of height $2\pi\mathrm{i}$, use the poles at $\mathrm{i}\theta$ and $2\pi\mathrm{i} - \mathrm{i}\theta$, and then let the horizontal width tend to infinity.

Writing $g_\theta(u) = (\cosh u - \cos\theta)^{-1}$, direct differentiation gives

$$\psi_\theta = -\sin\theta \, \partial_\theta g_\theta - \cos\theta \, g_\theta. \tag{4.5}$$

Differentiating (4.4) in θ and using (4.5), we obtain

$$\widehat{\psi_\theta}(\xi) = \frac{1}{\sqrt{2\pi}} \frac{2\pi\xi \cosh((\pi-\theta)\xi)}{\sinh(\pi\xi)}. \tag{4.6}$$

We also use

$$\int_0^\pi \mathrm{e}^{\xi\theta} (\sin\theta)^{d-2}\, \mathrm{d}\theta = \frac{\pi \mathrm{e}^{\pi\xi/2}\Gamma(d-1)}{2^{d-2}|\Gamma(d/2+\mathrm{i}\xi/2)|^2}. \tag{4.7}$$

Formula (4.7) follows from the classical beta integral by analytic continuation.

Substituting (4.6) into (4.2), and then using (4.7), gives

$$\widehat{\mathcal{K}_d^{(1)}}(\xi) = \frac{1}{\sqrt{2\pi}} \frac{c_d \pi^2 \xi \Gamma(d-1)}{2^{d-2}\sinh(\pi\xi/2)|\Gamma(d/2+\mathrm{i}\xi/2)|^2}. \tag{4.8}$$

The duplication formula implies

$$\frac{c_d \pi \Gamma(d-1)}{2^{d-2}} = \frac{1}{2}\Gamma(d/2)^2,$$

and

$$|\Gamma(1+\mathrm{i}\xi/2)|^2 = \frac{\pi\xi/2}{\sinh(\pi\xi/2)}.$$

Combining these identities with (4.8) proves (4.3). □

4.2 The Brownian representation

The kernel q_d in (2.2) satisfies

$$\int_0^\infty q_d(t)\, \mathrm{d}t = 1, \qquad q_d(t) \asymp t^{m-3/2} \quad (t \downarrow 0), \qquad q_d^{(j)}(t) = O_j(\mathrm{e}^{-2t}) \quad (t \to \infty). \tag{4.9}$$

In particular, $q_d \in L^1(0,\infty)$ for every odd $d \geq 3$.

Lemma 4.2 (Transfer function)**.** *For each $\xi \in \mathbb{R}$,*

$$Q_d(\xi) = \int_0^\infty q_d(t)\mathrm{e}^{-\mathrm{i}\xi t}\, \mathrm{d}t = \Gamma(d/2)\frac{\Gamma(1+\mathrm{i}\xi/2)}{\Gamma(d/2+\mathrm{i}\xi/2)}. \tag{4.10}$$

Consequently,

$$\widehat{\mathcal{K}_d^{(1)}}(\xi) = \frac{1}{\sqrt{2\pi}}|Q_d(\xi)|^2. \tag{4.11}$$

Proof. With $x = \mathrm{e}^{-2t}$,

$$\begin{aligned} Q_d(\xi) &= \frac{d-2}{2}\int_0^1 x^{\mathrm{i}\xi/2}(1-x)^{(d-4)/2}\, \mathrm{d}x \\ &= \frac{d-2}{2}\frac{\Gamma(1+\mathrm{i}\xi/2)\Gamma((d-2)/2)}{\Gamma(d/2+\mathrm{i}\xi/2)}. \end{aligned}$$

Since $(d-2)\Gamma((d-2)/2)/2 = \Gamma(d/2)$, this gives (4.10). Equation (4.11) follows from Lemma 4.1. □

Corollary 4.3 (Power spectrum in odd dimension)**.** *For $d = 2m + 1$,*

$$|Q_d(\xi)|^2 = 2^{d-2}\Gamma(d/2)^2 \frac{\xi \coth(\pi\xi/2)}{\prod_{j=0}^{m-1} (\xi^2 + (2j+1)^2)}. \tag{4.12}$$

Proof. Put $x = \xi/2$. Repeatedly using $\Gamma(z+1) = z\Gamma(z)$ gives

$$|\Gamma(m + 1/2 + \mathrm{i}x)|^2 = \prod_{j=0}^{m-1} \big(x^2 + (j+1/2)^2\big) |\Gamma(1/2 + \mathrm{i}x)|^2.$$

Together with

$$|\Gamma(1 + \mathrm{i}x)|^2 = \frac{\pi x}{\sinh(\pi x)}, \qquad |\Gamma(1/2 + \mathrm{i}x)|^2 = \frac{\pi}{\cosh(\pi x)},$$

this proves (4.12). □

Proof of Theorem A. For a two-sided Brownian path of at most polynomial growth, define

$$\mathcal{S}_d(B)_t \stackrel{\text{def}}{=} \int_0^\infty q_d(u)(B_{t-u} - B_{-u})\,\mathrm{d}u. \tag{4.13}$$

The integral exists pathwise by (4.9). The resulting function of t is continuous. Indeed, on a compact interval in t, split the integral at a large deterministic L, use uniform continuity of B on the part $u \le L$, and use exponential decay of q_d together with polynomial growth of B for the remaining part.

For $\varphi \in C_c^\infty(\mathbb{R})$, integration by parts in t shows that the generalized derivative of (4.13), tested against φ, is the white-noise pairing with $\widetilde{q}_d * \varphi$, where $\widetilde{q}_d(t) = q_d(-t)$. Its covariance distribution therefore has Fourier transform $(2\pi)^{-1/2}|Q_d|^2$. By (4.11), this is the covariance distribution of the generalized derivative of S.

It remains to recover the additive anchoring. If C_1, C_2 are the covariance kernels of S and $\mathcal{S}_d(B)$, respectively, then $D = C_1 - C_2$ satisfies $\partial_t \partial_s D = 0$ in $\mathcal{D}'(\mathbb{R}^2)$. Hence $D(t,s) = f(t) + g(s)$ distributionally. Both processes vanish at time zero, so $D(t,0) = D(0,s) = 0$, which implies $D = 0$. The finite-dimensional laws therefore agree. Since both processes have continuous versions, their laws agree on $C_{\mathrm{loc}}(\mathbb{R})$. □

Put

$$H_d(t) \stackrel{\text{def}}{=} 1 - K_d(t), \qquad t > 0, \tag{4.14}$$

and extend H_d by zero to $(-\infty, 0)$. Then

$$H_d'(t) = -q_d(t), \qquad H_d \in L^1(0,\infty) \cap L^2(0,\infty). \tag{4.15}$$

Proposition 4.4 (Brownian motion plus a stationary process)**.** *On the coupling of Theorem A, define*

$$Z_t \stackrel{\text{def}}{=} -\int_{-\infty}^t H_d(t-s)\,\mathrm{d}B_s. \tag{4.16}$$

Then Z has a stationary continuous Gaussian modification and

$$S_t = B_t + Z_t - Z_0, \qquad t \in \mathbb{R}. \tag{4.17}$$

Moreover,

$$\frac{S_t}{|t|} \longrightarrow 0 \qquad (|t| \to \infty) \tag{4.18}$$

almost surely. Consequently,

$$S_t - at \longrightarrow +\infty \quad (t \to -\infty), \qquad S_t - at \longrightarrow -\infty \quad (t \to +\infty). \tag{4.19}$$

Proof. The stochastic integral in (4.16) is well-defined by (4.15). The extension of H_d to $\mathbb{R}$ is bounded and has finite variation, so

$$\|H_d(\cdot + r) - H_d\|_{L^1(\mathbb{R})} \le |r| \operatorname{Var}(H_d).$$

It follows that

$$\mathbb{E}[(Z_{t+r} - Z_t)^2] = \|H_d(\cdot + r) - H_d\|^2_{L^2(\mathbb{R})} \le 2\|H_d\|_\infty \operatorname{Var}(H_d)|r|.$$

Gaussian fourth moments and Kolmogorov's criterion give a continuous modification. Stationarity follows directly from the white-noise representation.

Stochastic integration by parts on finite intervals, followed by an L^2 limit, gives

$$\int_0^\infty q_d(u)(B_{t-u} - B_{-u})\,\mathrm{d}u = B_t - \int_{-\infty}^t H_d(t-s)\,\mathrm{d}B_s + \int_{-\infty}^0 H_d(-s)\,\mathrm{d}B_s,$$

which is (4.17).

Brownian motion is sublinear. For the stationary continuous Gaussian process Z, the random variables $\sup_{t\in[n,n+1]} |Z_t|$ have a uniform sub-Gaussian tail. A union bound and the Borel–Cantelli lemma imply $\sup_{t\in[n,n+1]} |Z_t|/|n| \to 0$ in both time directions. Equation (4.17) gives (4.18), and (4.19) follows since $a > 0$. □

5 The conditional law of the future

5.1 The contribution from the past

For $\omega \in C_0(\mathbb{R}) \stackrel{\text{def}}{=} \{\omega \in C(\mathbb{R}) : \omega(0) = 0\}$, define

$$(\theta_t\omega)(s) \stackrel{\text{def}}{=} \omega(t+s) - \omega(t). \tag{5.1}$$

For a continuous path of at most polynomial growth, let

$$A_d(\omega) \stackrel{\text{def}}{=} \int_0^\infty q_d(u)\omega(-u)\,\mathrm{d}u. \tag{5.2}$$

Then the map in (4.13) satisfies

$$\mathcal{S}_d(\omega)_t = \omega(t) + A_d(\theta_t\omega) - A_d(\omega), \qquad \theta_t\mathcal{S}_d(\omega) = \mathcal{S}_d(\theta_t\omega). \tag{5.3}$$

The following elementary estimate will be used for the Brownian path in Section 6.

Lemma 5.1 (Sublinear growth)**.** *Suppose that ω is continuous, has at most polynomial growth, and $\omega(t)/|t| \to 0$ as $|t| \to \infty$. Then*

$$\frac{\mathcal{S}_d(\omega)_t}{|t|} \longrightarrow 0 \qquad (|t| \to \infty). \tag{5.4}$$

Proof. Put $T = |t|$ and split the integral in (4.13) at $u = T/2$. Given $\varepsilon > 0$, choose M such that $|\omega(x)| \le \varepsilon|x|$ for $|x| \ge M$. If $T \ge 2M$ and $0 \le u \le T/2$, then $|t-u| \ge T/2$, so

$$|\omega(t-u)| \le \frac{3}{2}\varepsilon T, \qquad |\omega(-u)| \le \varepsilon u + C_M, \qquad C_M \stackrel{\text{def}}{=} \sup_{|x|\le M} |\omega(x)|.$$

Since q_d integrates to one, the contribution of $[0, T/2]$, divided by T, is at most $2\varepsilon + C_M/T$. On $u > T/2$, polynomial growth of ω and exponential decay of q_d give a bound of the form $C(1+T)^N \mathrm{e}^{-T}$. Letting $T \to \infty$ and then $\varepsilon \downarrow 0$ proves (5.4). □

Fix $\lambda \in (0, 2)$. We introduce a space in which to record the function appearing in the second term on the right-hand side of (1.6).

Definition 5.2. Let $\mathcal{E}_{\lambda,m}$ be the set of functions $M \in C([0,\infty))$ such that $M|_{[1,\infty)} \in H^{m+1}_{\mathrm{loc}}([1,\infty))$ and

$$\|M\|_{\mathcal{E}_{\lambda,m}} \stackrel{\text{def}}{=} \sup_{0\le r\le 1} |M(r)| + \sum_{j=0}^{m+1} \left(\int_1^\infty \mathrm{e}^{2\lambda r} |M^{(j)}(r)|^2 \, \mathrm{d}r \right)^{1/2} < \infty. \tag{5.5}$$

Weak derivatives on $[1,\infty)$ are represented by their continuous Sobolev representatives whenever point evaluation is used.

The space $\mathcal{E}_{\lambda,m}$ is a separable Banach space. To see this, let w be smooth and positive on $[1,\infty)$ and equal to $\mathrm{e}^{\lambda r}$ for $r \ge 2$. By Leibniz' rule, the tail part of (5.5) is equivalent to $\|wM\|_{H^{m+1}(1,\infty)}$. Hence $\mathcal{E}_{\lambda,m}$ is isomorphic to a closed subspace of $C([0,1]) \times H^{m+1}(1,\infty)$, with the two components identified at the trace $r = 1$.

We will repeatedly use the estimate

$$|M^{(k)}(u)| \le C_{m,\lambda} \mathrm{e}^{-\lambda u} \|M\|_{\mathcal{E}_{\lambda,m}}, \qquad u \ge 2, \quad 0 \le k \le m. \tag{5.6}$$

It follows from the one-dimensional embedding $H^1 \hookrightarrow C^0$ applied to $\mathrm{e}^{\lambda r} M^{(k)}(r)$ on $[u-1, u+1]$.

For a path ω as above, define

$$M_t(\omega)(r) \stackrel{\text{def}}{=} \int_{-\infty}^0 q_d(r-s)(\omega(t+s) - \omega(t)) \, \mathrm{d}s, \qquad r \ge 0. \tag{5.7}$$

When $r = 0$, the only singular case is $d = 3$, in which $q_d(-s) = O(|s|^{-1/2})$ as $s \uparrow 0$. Hence (5.7) is well-defined for every locally Hölder path of positive exponent.

Lemma 5.3 (Continuity of the past contribution)**.** *Suppose that $\omega \in C(\mathbb{R})$ has at most polynomial growth and is locally Hölder continuous with some positive exponent on each compact interval. Then $M_t(\omega) \in \mathcal{E}_{\lambda,m}$ for every $t \in \mathbb{R}$, the map*

$$t \longmapsto M_t(\omega) \tag{5.8}$$

is continuous from $\mathbb{R}$ to $\mathcal{E}_{\lambda,m}$, and

$$M_t(\omega) = M_0(\theta_t \omega). \tag{5.9}$$

The conclusion holds almost surely for a two-sided Brownian motion and for the path $\widetilde{B}$ introduced in Section 3.

Proof. Identity (5.9) is immediate. Fix a compact interval J of values of t. For $r \in [0,1]$, split (5.7) at $s = -\delta$. Uniform local Hölder continuity gives, for some $\alpha > 0$,

$$\sup_{t\in J}\sup_{0\le r\le 1}\int_{-\delta}^{0} q_d(r-s)|\omega(t+s)-\omega(t)|\,\mathrm{d}s \\ \le C_J\int_0^{\delta}\overline{q}_d(v)v^{\alpha}\,\mathrm{d}v \longrightarrow 0 \qquad (\delta\downarrow 0),$$

where $\overline{q}_d(v) = \sup_{0\le r\le 1} q_d(r+v)$. For $0 < v \le 1$, $\overline{q}_d(v) \le C_d(1+v^{-1/2})$. On $(-\infty,-\delta]$, continuity in (t,r) and exponential decay of q_d give uniform convergence after truncating the tail. It follows that $t \mapsto M_t(\omega)|_{[0,1]}$ is continuous in the supremum norm.

For $r \ge 1$ and $0 \le j \le m+1$, differentiation under the integral is justified by

$$|q_d^{(j)}(x)| \le C_{d,j}\mathrm{e}^{-2x}, \qquad x \ge 1,$$

and gives

$$\partial_r^j M_t(\omega)(r) = \int_{-\infty}^{0} q_d^{(j)}(r-s)(\omega(t+s)-\omega(t))\,\mathrm{d}s. \tag{5.10}$$

Uniformly for $t \in J$, the absolute value of (5.10) is at most $C_J\mathrm{e}^{-2r}$, where

$$C_J \le C\sup_{t\in J}\int_{-\infty}^{0}\mathrm{e}^{2s}|\omega(t+s)-\omega(t)|\,\mathrm{d}s < \infty.$$

Since $\lambda < 2$, all terms in (5.5) are finite. Dominated convergence gives their continuity in t. □

Let

$$\mathcal{F}_t \stackrel{\text{def}}{=} \sigma(B_s : s \le t), \qquad t \ge 0, \tag{5.11}$$

where the entire path on $(-\infty, 0]$ is included in $\mathcal{F}_0$.

Proposition 5.4 (Conditional law after a stopping time)**.** *For the Brownian motion in Theorem A,*

$$M_t(r) = -\int_{-\infty}^{t} H_d(r+t-s)\,\mathrm{d}B_s, \qquad r \ge 0. \tag{5.12}$$

The pathwise and stochastic definitions agree almost surely. If $\tau \ge 0$ is an $(\mathcal{F}_t)$-stopping time and

$$W_t \stackrel{\text{def}}{=} B_{\tau+t} - B_\tau,$$

then W is independent of $\mathcal{F}_\tau$ and

$$S_{\tau+t} - S_\tau = \int_0^t K_d(t-s)\,\mathrm{d}W_s + M_\tau(t) - M_\tau(0), \qquad t \ge 0. \tag{5.13}$$

Proof. Fix $r \ge 0$ and truncate (5.7) to $s \in [-R, -\varepsilon]$. Since $\partial_s H_d(r-s) = q_d(r-s)$, stochastic integration by parts gives

$$\int_{-R}^{-\varepsilon} q_d(r-s)(B_{t+s}-B_t)\,\mathrm{d}s \\ = \big[H_d(r-s)(B_{t+s}-B_t)\big]_{-R}^{-\varepsilon} - \int_{t-R}^{t-\varepsilon} H_d(r+t-v)\,\mathrm{d}B_v.$$

The boundary term at $-R$ tends to zero in L^2, and almost surely along integers, by exponential decay. The term at $-\varepsilon$ tends to zero almost surely because Brownian paths are locally Hölder continuous with every exponent less than $1/2$, while H_d is bounded. The stochastic integrals converge in L^2 since $H_d \in L^2(0,\infty)$. This proves (5.12).

Starting from (4.17), split the stochastic integral defining $Z_{\tau+t}$ at τ. The part before τ is $M_\tau(t)$ and the part after τ is $-\int_0^t H_d(t-s)\,\mathrm{d}W_s$. Since $W_t = \int_0^t 1\,\mathrm{d}W_s$, these terms combine into $\int_0^t K_d(t-s)\,\mathrm{d}W_s$. Subtracting $Z_\tau = M_\tau(0)$ proves (5.13). □

5.2 Inversion of the kernel on the half-line

We first record two facts concerning zero extensions at the origin. The first is the standard trace theorem for integer order Sobolev spaces.

Lemma 5.5 (Zero extension at integer order)**.** *Let $k \in \mathbb{N}$. Suppose that $f \in H^k(0,\infty)$ and*

$$f^{(j)}(0) = 0, \qquad 0 \le j \le k-1. \tag{5.14}$$

Then the extension of f by zero to $(-\infty,0)$ belongs to $H^k(\mathbb{R})$, with norm at most $C_k\|f\|_{H^k(0,\infty)}$.

Proof. For smooth f, the distributional derivatives of the zero extension contain boundary delta terms whose coefficients are the traces in (5.14). Under the stated assumptions these terms vanish through order k. Approximation by smooth functions preserving the traces gives the general case; see [1, Chapter 7]. □

Define

$$E_m(t) \stackrel{\text{def}}{=} \mathrm{e}^{-t}\sum_{\ell=0}^{m-1}\frac{t^\ell}{\ell!}, \qquad F_m(t) \stackrel{\text{def}}{=} K_d(t) - 1 + E_m(t), \qquad t \ge 0. \tag{5.15}$$

Lemma 5.6 (A half-integer trace estimate)**.** *The extension of F_m by zero to $(-\infty,0)$ belongs to $H^\rho(\mathbb{R})$, where $\rho = m - 1/2$.*

Proof. Since $K_d(t) = (1-\mathrm{e}^{-2t})^{m-1/2}$,

$$F_m^{(j)}(0) = 0 \quad (0 \le j \le m-1), \qquad F_m(t) = 2^{m-1/2}t^{m-1/2} + O(t^m) \quad (t \downarrow 0). \tag{5.16}$$

Both $K_d(t)-1$ and $E_m(t)$ decay exponentially, together with all of their derivatives, as $t \to \infty$. The Fourier characterisation of integer derivatives gives

$$\|f\|^2_{H^{m-1/2}(\mathbb{R})} \asymp_m \|f\|^2_{L^2(\mathbb{R})} + \|f^{(m-1)}\|^2_{H^{1/2}(\mathbb{R})} \tag{5.17}$$

whenever the right-hand side is finite. The vanishing traces in (5.16) imply that the $(m-1)$st derivative of the zero extension is the zero extension of $F_m^{(m-1)}$.

Near zero, $F_m^{(m-1)}$ is a constant multiple of $t_+^{1/2}$ plus a Lipschitz function which vanishes at zero. The local Gagliardo seminorm of the singular term is finite because

$$\int_0^1\int_0^1 \frac{|\sqrt{t}-\sqrt{s}|^2}{|t-s|^2}\,\mathrm{d}t\,\mathrm{d}s = \int_0^1\int_0^1 \frac{\mathrm{d}t\,\mathrm{d}s}{(\sqrt{t}+\sqrt{s})^2} < \infty, \tag{5.18}$$

and the contribution across the boundary is

$$\int_0^1 t\left(\int_{-\infty}^0 \frac{\mathrm{d}s}{(t-s)^2}\right)\mathrm{d}t < \infty.$$

Away from zero the function is smooth and exponentially decaying. Thus the zero extension of $F_m^{(m-1)}$ belongs to $H^{1/2}(\mathbb{R})$, and (5.17) proves the lemma. □

For $\Re z > 0$, the Laplace transform of q_d is

$$\mathcal{Q}_d(z) \stackrel{\text{def}}{=} \int_0^\infty \mathrm{e}^{-zt} q_d(t)\,\mathrm{d}t = \Gamma(d/2)\frac{\Gamma(1+z/2)}{\Gamma(d/2+z/2)}. \tag{5.19}$$

Proposition 5.7 (Paley–Wiener in Sobolev form)**.** *Let $s \geq 0$. If $f \in H^s(\mathbb{R})$ and $\operatorname{supp} f \subset [0,\infty)$, then its Laplace transform $F = \mathcal{L}f$ satisfies*

$$(1+z)^s F(z) \in H^2(\mathbb{C}_+), \qquad \|(1+\cdot)^s F\|_{H^2} \asymp_s \|f\|_{H^s(\mathbb{R})}. \tag{5.20}$$

Conversely, the inverse Paley–Wiener transform of a function in $H^2(\mathbb{C}_+)$ is supported in $[0,\infty)$.

Proof. The Paley–Wiener theorem identifies $L^2(0,\infty)$ with $H^2(\mathbb{C}_+)$. The analytic function $(1+z)^s F(z)$ belongs to the Smirnov class and has non-tangential boundary value $(1+\mathrm{i}\xi)^s \widehat{f}(\xi)$, up to the Fourier normalisation. Its boundary modulus is square integrable exactly when $f \in H^s(\mathbb{R})$. The Smirnov maximum theorem gives (5.20); see [5, Chapters 2 and 4]. □

Proposition 5.8 (Inverse of q_d)**.** *Suppose that $f \in H^\rho(\mathbb{R})$ and $\operatorname{supp} f \subset [0,\infty)$. There is a unique $g \in L^2(0,\infty)$ such that*

$$q_d * g = f \quad \text{in } L^2(0,\infty), \tag{5.21}$$

and

$$\|g\|_{L^2(0,\infty)} \leq C_d \|f\|_{H^\rho(\mathbb{R})}. \tag{5.22}$$

Proof. Define on $\mathbb{C}_+$, using the principal branch,

$$\mathcal{M}_d(z) \stackrel{\text{def}}{=} \frac{1}{(1+z)^\rho \mathcal{Q}_d(z)} = \frac{\Gamma(d/2+z/2)}{\Gamma(d/2)(1+z)^\rho \Gamma(1+z/2)}. \tag{5.23}$$

The Gamma function has no zeros. The uniform Gamma-ratio estimate obtained from Stirling's formula gives

$$\frac{\Gamma(w+d/2)}{\Gamma(w+1)} = w^\rho\big(1+O_d(|w|^{-1})\big), \qquad |w| \to \infty, \quad |\arg w| \leq \frac{\pi}{2}. \tag{5.24}$$

Taking $w = z/2$ shows that $\mathcal{M}_d$ is bounded for large $|z|$ with $\Re z \geq 0$. It is bounded on the remaining compact set since all possible poles lie strictly to the left. Hence $\mathcal{M}_d \in H^\infty(\mathbb{C}_+)$.

Let $F = \mathcal{L}f$. By Proposition 5.7, $(1+z)^\rho F(z) \in H^2(\mathbb{C}_+)$. Therefore

$$G(z) \stackrel{\text{def}}{=} \mathcal{M}_d(z)(1+z)^\rho F(z) = \frac{F(z)}{\mathcal{Q}_d(z)}$$

belongs to $H^2(\mathbb{C}_+)$. Let $g \in L^2(0,\infty)$ be its Paley–Wiener inverse. Then

$$\mathcal{L}(q_d * g) = \mathcal{Q}_d G = F,$$

which proves (5.21). The multiplier bound gives (5.22). Uniqueness follows because $\mathcal{Q}_d$ has no zeros on $\mathbb{C}_+$. □

5.3 A comparison estimate for conditional laws

For $M \in \mathcal{E}_{\lambda,m}$ and $u \geq 0$, let P_u^M be the law on $C([0,\infty))$ of

$$V_t^{M,u} \stackrel{\text{def}}{=} \int_0^{u+t} K_d(u+t-s)\,\mathrm{d}W_s + M(u+t) - M(0), \qquad t \geq 0, \tag{5.25}$$

where W is a standard Brownian motion. By Proposition 5.4, this is the conditional law of the path beginning u units after a stopping time, when the contribution from the past is M.

Proposition 5.9 (Comparison of conditional laws)**.** *There is $C_{d,\lambda} < \infty$ such that, for $u \geq 3$ and $M, N \in \mathcal{E}_{\lambda,m}$,*

$$d_{\mathrm{TV}}(P_u^M, P_u^N) \leq C_{d,\lambda}\left(\frac{|M(0)-N(0)|}{\sqrt{u}} + \mathrm{e}^{-\lambda u}\|M-N\|_{\mathcal{E}_{\lambda,m}}\right). \tag{5.26}$$

Proof. Put $D = M - N$, $c = -D(0)$, and

$$f(t) \stackrel{\text{def}}{=} D(u+t) - D(0) = c + D(u+t), \qquad t \geq 0. \tag{5.27}$$

We will construct $g \in L^2(0,\infty)$ such that

$$\int_0^{u+t} K_d(u+t-s)g(s)\,\mathrm{d}s = f(t), \qquad t \geq 0, \tag{5.28}$$

and

$$\|g\|^2_{L^2(0,\infty)} \leq C_{d,\lambda}\left(\frac{c^2}{u} + \mathrm{e}^{-2\lambda u}\|D\|^2_{\mathcal{E}_{\lambda,m}}\right). \tag{5.29}$$

Let

$$\mathcal{V} \stackrel{\text{def}}{=} \left\{\phi \in C_c^\infty((1,2)) : \int_1^2 \phi(v)\,\mathrm{d}v = 0\right\} \tag{5.30}$$

and, for $0 \leq k \leq m$, define

$$L_k(\phi) \stackrel{\text{def}}{=} \int_1^2 K_d^{(k)}(v)\phi(v)\,\mathrm{d}v. \tag{5.31}$$

The functionals $L_0, \ldots, L_m$ are linearly independent on $\mathcal{V}$. Indeed, a dependence would imply that a nonzero linear combination of $K_d, K_d', \ldots, K_d^{(m)}$ is constant on $(1,2)$. Since

$$K_d(t) = (1-\mathrm{e}^{-2t})^{m-1/2} = \sum_{n=0}^{\infty}(-1)^n\binom{m-1/2}{n}\mathrm{e}^{-2nt}, \tag{5.32}$$

and every coefficient with $n \geq 1$ is nonzero, the nonconstant exponential coefficients would give a polynomial of degree at most m which vanishes at $-2n$ for every $n \geq 1$. This is impossible.

Choose $\phi_0, \ldots, \phi_m \in \mathcal{V}$ such that the matrix $(L_k(\phi_j))_{0\leq k,j\leq m}$ is invertible. On $[0,u]$, set

$$g_0(s) \stackrel{\text{def}}{=} \frac{c}{u} + \sum_{j=0}^{m} a_j\phi_j(u-s). \tag{5.33}$$

The second term is supported in $[u-2, u-1]$ and has integral zero. Choose the coefficients a_j so that

$$\int_0^u K_d^{(k)}(u-s)g_0(s)\,\mathrm{d}s = f^{(k)}(0), \qquad 0 \leq k \leq m. \tag{5.34}$$

All traces in (5.34) are well-defined. For $k = 0$, the contribution of the constant part in (5.33) is $cu^{-1}\int_0^u K_d(v)\,\mathrm{d}v$. For $1 \le k \le m$, it is $cu^{-1}K_d^{(k-1)}(u)$, since

$$K_d^{(j)}(0) = 0, \qquad 0 \le j \le m-1. \tag{5.35}$$

Also,

$$u - \int_0^u K_d(v)\,\mathrm{d}v = \int_0^u H_d(v)\,\mathrm{d}v = O_d(1). \tag{5.36}$$

The invertibility of the fixed matrix, (5.6), and exponential decay of the positive-order derivatives of K_d imply

$$\sum_{j=0}^{m} |a_j| \le C_{d,\lambda}\left(\frac{|c|}{u} + \mathrm{e}^{-\lambda u}\|D\|_{\mathcal{E}_{\lambda,m}}\right). \tag{5.37}$$

Consequently,

$$\|g_0\|^2_{L^2(0,u)} \le C_{d,\lambda}\left(\frac{c^2}{u} + \mathrm{e}^{-2\lambda u}\|D\|^2_{\mathcal{E}_{\lambda,m}}\right). \tag{5.38}$$

For $t \ge 0$, define

$$r(t) \stackrel{\text{def}}{=} f(t) - \int_0^u K_d(u+t-s)g_0(s)\,\mathrm{d}s. \tag{5.39}$$

By (5.34),

$$r^{(k)}(0) = 0, \qquad 0 \le k \le m. \tag{5.40}$$

Moreover, $\int_0^u g_0 = c$, $D(u+t) \to 0$, and $K_d(t) \to 1$ exponentially, so $r(t) \to 0$ exponentially. Differentiating (5.39) gives

$$r'(t) = D'(u+t) + \frac{c}{u}\big(K_d(t) - K_d(u+t)\big) - \sum_{j=0}^{m} a_j G_j(t), \tag{5.41}$$

where

$$G_j(t) \stackrel{\text{def}}{=} \int_1^2 q_d(t+v)\phi_j(v)\,\mathrm{d}v. \tag{5.42}$$

Using (5.15), write

$$r'(t) = \frac{c}{u}F_m(t) + A_u(t), \tag{5.43}$$

where

$$A_u(t) = D'(u+t) + \frac{c}{u}\big(H_d(u+t) - E_m(t)\big) - \sum_{j=0}^{m} a_j G_j(t). \tag{5.44}$$

The function $D'(u+\cdot)$ belongs to $H^m(0,\infty)$, and every other term in (5.44) is smooth and exponentially decaying. Hence $A_u \in H^m(0,\infty)$. By (5.40) and the vanishing traces of F_m,

$$A_u^{(j)}(0) = 0, \qquad 0 \le j \le m-1. \tag{5.45}$$

Using (5.37) and exponential decay, we obtain

$$\|A_u\|_{H^m(0,\infty)} \le C_{d,\lambda}\left(\frac{|c|}{u} + \mathrm{e}^{-\lambda u}\|D\|_{\mathcal{E}_{\lambda,m}}\right). \tag{5.46}$$

By Lemma 5.5, the zero extension of A_u lies in $H^m(\mathbb{R})$. By Lemma 5.6, the zero extension of F_m lies in $H^\rho(\mathbb{R})$. Since $m > \rho$, (5.43) gives

$$\|r'\|_{H^\rho(\mathbb{R})} \le C_{d,\lambda}\left(\frac{|c|}{u} + \mathrm{e}^{-\lambda u}\|D\|_{\mathcal{E}_{\lambda,m}}\right), \tag{5.47}$$

where r' is extended by zero to $(-\infty, 0)$.

Apply Proposition 5.8 to r'. There is $g_1 \in L^2(0,\infty)$ such that

$$q_d * g_1 = r' \tag{5.48}$$

and

$$\|g_1\|_{L^2(0,\infty)} \le C_{d,\lambda}\left(\frac{|c|}{u} + \mathrm{e}^{-\lambda u}\|D\|_{\mathcal{E}_{\lambda,m}}\right). \tag{5.49}$$

Since $K_d(0) = 0$ and $K_d' = q_d$, the function $K_d * g_1$ has derivative r' and trace zero at the origin. By (5.40),

$$K_d * g_1 = r. \tag{5.50}$$

Finally, set

$$g(s) \stackrel{\text{def}}{=} \begin{cases} g_0(s), & 0 \le s \le u, \\ g_1(s-u), & s > u. \end{cases} \tag{5.51}$$

Equations (5.39) and (5.50) give (5.28), while (5.38) and (5.49) give (5.29).

Translate Wiener measure by the Cameron–Martin path $t \mapsto \int_0^t g(s)\,\mathrm{d}s$. Under this shift, the output in (5.25) with N in place of M is translated exactly into the output with M. The relative entropy of the shifted Wiener measure with respect to Wiener measure on $C([0,\infty))$ is $\frac{1}{2}\|g\|^2_{L^2(0,\infty)}$; see Lemma A.1. Relative entropy decreases under measurable maps, and Pinsker's inequality gives

$$d_{\mathrm{TV}}(P_u^M, P_u^N) \le \frac{1}{2}\|g\|_{L^2(0,\infty)}.$$

Together with (5.29), this proves (5.26). □

6 Quantum cones in arbitrary odd dimension

We now prove Theorems B and C. We continue to write $a = Q - \beta > 0$. All hitting times in this section refer to the continuous function $t \mapsto S_t - at$ or to the analogous function constructed from another path.

6.1 Tightness of the recentred process

Let $\widetilde{B}$ be the two-sided path from Section 3, and set

$$\widetilde{S} \stackrel{\text{def}}{=} \mathcal{S}_d(\widetilde{B}), \qquad \widetilde{\sigma}_b \stackrel{\text{def}}{=} \inf\{t \in \mathbb{R} : \widetilde{S}_t - at = -b\}. \tag{6.1}$$

By Lemma 5.1 and (3.12),

$$\widetilde{S}_t - at \longrightarrow +\infty \quad (t \to -\infty), \qquad \widetilde{S}_t - at \longrightarrow -\infty \quad (t \to +\infty). \tag{6.2}$$

In particular, $\widetilde{\sigma}_b$ is finite. The second identity in (5.3) and the recentering identity (3.11) imply

$$\theta_{\tau_b}\widetilde{S} \stackrel{\text{law}}{=} \widetilde{S}. \tag{6.3}$$

Lemma 6.1 (Comparison of the two hitting times)**.** *The family*

$$(\widetilde{\sigma}_b - \tau_b)_{b>0} \tag{6.4}$$

is tight.

Proof. At time τ_b, define

$$\Delta_b \stackrel{\text{def}}{=} \widetilde{S}_{\tau_b} - a\tau_b + b. \tag{6.5}$$

Since $\widetilde{B}_{\tau_b} - a\tau_b = -b$, the first identity in (5.3) gives

$$\Delta_b = A_d(\theta_{\tau_b}\widetilde{B}) - A_d(\widetilde{B}). \tag{6.6}$$

By (3.11), both terms on the right have the same fixed marginal law. Hence $(\Delta_b)_{b>0}$ is tight.

For a path ω satisfying the two limits in (6.2), put

$$\kappa(\omega, c) \stackrel{\text{def}}{=} \inf\{r \in \mathbb{R} : \omega(r) - ar = -c\}. \tag{6.7}$$

The identity

$$\widetilde{S}_{\tau_b+r} - a(\tau_b + r) + b = (\theta_{\tau_b}\widetilde{S})(r) - ar + \Delta_b$$

shows that

$$\widetilde{\sigma}_b - \tau_b = \kappa(\theta_{\tau_b}\widetilde{S}, \Delta_b). \tag{6.8}$$

For $C > 0$, let $R_C(\omega)$ be the least integer $R \geq 1$ such that

$$\omega(r) - ar > C \quad (r \leq -R), \qquad \omega(r) - ar < -C \quad (r \geq R). \tag{6.9}$$

The quantity $R_C(\omega)$ is finite for every path under consideration, and

$$|\kappa(\omega, c)| \leq R_C(\omega) \qquad \text{whenever } |c| \leq C. \tag{6.10}$$

The law of $\theta_{\tau_b}\widetilde{S}$ does not depend on b by (6.3). Combining this fact with the tightness of Δ_b and (6.10) proves the lemma. □

The same argument also controls changes in the hitting level.

Lemma 6.2 (Bounded changes of level)**.** *For each $C < \infty$, the two families*

$$\{\tau_{b+c} - \tau_b : b > C + 1,\ |c| \leq C\} \quad \textit{and} \quad \{\widetilde{\sigma}_{b+c} - \widetilde{\sigma}_b : b > C + 1,\ |c| \leq C\} \tag{6.11}$$

are tight.

Proof. After time τ_b, the relative time at which the Brownian path first reaches the additional level $-c$ is

$$\inf\{r \in \mathbb{R} : (\theta_{\tau_b}\widetilde{B})(r) - ar = -c\}.$$

Its absolute value is bounded by $R_C(\theta_{\tau_b}\widetilde{B})$, with R_C defined as in (6.9). The law of this last random variable does not depend on b by (3.11). This proves the first assertion. The second follows from Lemma 6.1 and the triangle inequality. □

We next compare the path $\widetilde{B}$ with an ordinary two-sided Brownian motion. Couple B and $\widetilde{B}$ so that

$$B_t = \widetilde{B}_t, \qquad t \geq 0. \tag{6.12}$$

Let $S = \mathcal{S}_d(B)$ and retain the notation $\widetilde{S} = \mathcal{S}_d(\widetilde{B})$.

Lemma 6.3 (Difference between the two spherical average processes)**.** *There is a finite random variable*

$$C \stackrel{\text{def}}{=} A_d(\widetilde{B}) - A_d(B) \tag{6.13}$$

such that

$$S_t - \widetilde{S}_t \longrightarrow C \qquad (t \to \infty) \tag{6.14}$$

almost surely. Furthermore,

$$\|M_t(B) - M_t(\widetilde{B})\|_{\mathcal{E}_{\lambda,m}} \longrightarrow 0 \qquad (t \to \infty) \tag{6.15}$$

almost surely.

Proof. By (5.3) and (6.12),

$$S_t - \widetilde{S}_t = A_d(\theta_t B) - A_d(\theta_t \widetilde{B}) + C. \tag{6.16}$$

The first two terms on the right differ by

$$\int_t^\infty q_d(u)(B_{t-u} - \widetilde{B}_{t-u})\,\mathrm{d}u. \tag{6.17}$$

The negative-time paths have at most polynomial growth, whereas q_d decays exponentially. The expression in (6.17) therefore tends to zero, proving (6.14).

Changing variables in (5.7) gives, for $t \geq 0$ and $r \geq 0$,

$$\big(M_t(B) - M_t(\widetilde{B})\big)(r) = \int_{-\infty}^0 q_d(r + t - v)(B_v - \widetilde{B}_v)\,\mathrm{d}v. \tag{6.18}$$

For t large, every argument of q_d in (6.18) is bounded away from zero. For $0 \leq j \leq m+1$ and $x \geq 1$,

$$|q_d^{(j)}(x)| \leq C_{d,j}\mathrm{e}^{-2x}. \tag{6.19}$$

It follows from (6.18) and (6.19) that the $C([0,1])$ part and every weighted derivative term in (5.5) are at most a deterministic multiple of e^{-2t} times

$$\int_{-\infty}^0 \mathrm{e}^{2v}|B_v - \widetilde{B}_v|\,\mathrm{d}v,$$

which is finite almost surely. This proves (6.15). □

Recall the ordinary hitting time

$$\sigma_b = \inf\{t \in \mathbb{R} : S_t - at = -b\}. \tag{6.20}$$

Lemma 6.4 (Comparison with the Brownian hitting time)**.** *Under the coupling* (6.12)*, the family*

$$(\sigma_b - \tau_b)_{b\to\infty} \tag{6.21}$$

is tight.

Proof. We first compare σ_b with $\widetilde{\sigma}_b$. Fix $\varepsilon > 0$. By Lemma 6.3, we can first choose $C_0, T < \infty$ and then b_0 sufficiently large so that, with probability at least $1 - \varepsilon$ for $b \geq b_0$,

$$|C| \leq C_0, \qquad \sup_{t \geq T} |S_t - \widetilde{S}_t - C| \leq 1, \tag{6.22}$$

and all hitting times appearing below are at least T. On this event,

$$\widetilde{\sigma}_{b+C-1} \leq \sigma_b \leq \widetilde{\sigma}_{b+C+1}. \tag{6.23}$$

Indeed, before the time on the left, $\widetilde{S}_t - at > -b - C + 1$, hence $S_t - at > -b$ by (6.22); at the time on the right, $\widetilde{S}_t - at = -b - C - 1$, hence $S_t - at \leq -b$.

Lemma 6.2 and (6.23) imply tightness of $\sigma_b - \widetilde{\sigma}_b$. Combining this with Lemma 6.1 proves (6.21). □

The global hitting time σ_b need not be a stopping time for the filtration (5.11). For the strong Markov arguments below, define

$$\sigma_b^+ \stackrel{\text{def}}{=} \inf\{t \geq 0 : S_t - at = -b\}. \tag{6.24}$$

By (4.19), the random variable $\inf_{t \leq 0}(S_t - at)$ is finite. Consequently,

$$\mathbb{P}[\sigma_b \neq \sigma_b^+] \longrightarrow 0 \qquad (b \to \infty). \tag{6.25}$$

Proposition 6.5 (Tightness at the recentering time)**.** *The two families*

$$(M_{\sigma_b}(B))_{b \to \infty} \quad \text{and} \quad (M_{\sigma_b^+}(B))_{b \to \infty} \tag{6.26}$$

are tight in $\mathcal{E}_{\lambda,m}$. For every $L < \infty$, the families

$$\big((S_{\sigma_b + t} - S_{\sigma_b})_{0 \leq t \leq L}\big)_{b \to \infty} \quad \text{and} \quad \big((S_{\sigma_b^+ + t} - S_{\sigma_b^+})_{0 \leq t \leq L}\big)_{b \to \infty} \tag{6.27}$$

are tight in $C([0, L])$.

Proof. By (5.9),

$$M_{\tau_b + r}(\widetilde{B}) = M_r(\theta_{\tau_b} \widetilde{B}). \tag{6.28}$$

For each fixed L, the $C([-L, L], \mathcal{E}_{\lambda,m})$-valued process on the right-hand side of (6.28) has a law independent of b, by (3.11) and Lemma 5.3. This fixed probability law is tight. If $\mathcal{K} \subset C([-L, L], \mathcal{E}_{\lambda,m})$ is compact, then

$$\{f(r) : f \in \mathcal{K},\ |r| \leq L\} \tag{6.29}$$

is compact in $\mathcal{E}_{\lambda,m}$, since it is the image of the compact set $\mathcal{K} \times [-L, L]$ under the evaluation map.

Lemma 6.4 allows us to choose L so that $|\sigma_b - \tau_b| \leq L$ with arbitrarily high probability, uniformly for all sufficiently large b. On the intersection with a compact-path event as above, $M_{\sigma_b}(\widetilde{B})$ belongs to a fixed compact subset of $\mathcal{E}_{\lambda,m}$. Moreover, $\sigma_b \to \infty$ in probability by (4.19); hence (6.15) implies

$$\|M_{\sigma_b}(B) - M_{\sigma_b}(\widetilde{B})\|_{\mathcal{E}_{\lambda,m}} \longrightarrow 0 \tag{6.30}$$

in probability. This proves tightness of the first family in (6.26). The second follows from (6.25).

The time σ_b^+ is a stopping time for (5.11). Conditional on $M_{\sigma_b^+}$, Proposition 5.4 gives

$$S_{\sigma_b^+ + t} - S_{\sigma_b^+} = \int_0^t K_d(t - s)\, \mathrm{d}W_s + M_{\sigma_b^+}(t) - M_{\sigma_b^+}(0), \qquad t \geq 0. \tag{6.31}$$

The first term has a fixed continuous law, and the restriction map from $\mathcal{E}_{\lambda,m}$ to $C([0, L])$ is continuous. Tightness of $M_{\sigma_b^+}$ therefore gives tightness of the second family in (6.27). The first follows from (6.25). □

6.2 Convergence of the recentred process

In the remainder of the proof we first use positive hitting times and return to the global hitting times at the end. Let R be an independent copy of S, represented using an independent two-sided Brownian motion, and set

$$\rho_b \overset{\text{def}}{=} \inf\{t \geq 0 : R_t - at = -b\}, \qquad R_{b,t} \overset{\text{def}}{=} R_{\rho_b+t} - R_{\rho_b}. \tag{6.32}$$

Let N have the stationary law of M_0, independently of all other random variables.

Proposition 6.6 (Comparison after the hitting time)**.** *One has*

$$\lim_{u\to\infty} \limsup_{b\to\infty} d_{\mathrm{TV}}\left(\mathrm{Law}((R_{b,u+t})_{t\geq 0}), \mathrm{Law}((S_{u+t})_{t\geq 0})\right) = 0. \tag{6.33}$$

Proof. Conditional on the Brownian path up to time ρ_b, Proposition 5.4 gives

$$\mathrm{Law}((R_{b,u+t})_{t\geq 0} \mid \mathcal{F}_{\rho_b}) = P_u^{M_{\rho_b}}. \tag{6.34}$$

By stationarity of the Brownian increments and (5.9),

$$\mathrm{Law}((S_{u+t})_{t\geq 0}) = \int_{\mathcal{E}_{\lambda,m}} P_u^M \ \mathrm{Law}(N)(\,\mathrm{d}M). \tag{6.35}$$

Proposition 6.5 implies that $(M_{\rho_b})_{b\to\infty}$ is tight in $\mathcal{E}_{\lambda,m}$. The random variable N has one fixed probability law on the same Polish space. Given $\varepsilon > 0$, choose a compact set $\mathcal{K}_\varepsilon \subset \mathcal{E}_{\lambda,m}$ such that

$$\limsup_{b\to\infty} \mathbb{P}[M_{\rho_b} \notin \mathcal{K}_\varepsilon] \leq \varepsilon, \qquad \mathbb{P}[N \notin \mathcal{K}_\varepsilon] \leq \varepsilon. \tag{6.36}$$

Convexity of total variation distance and Proposition 5.9 yield

$$\begin{aligned} &\limsup_{b\to\infty} d_{\mathrm{TV}}\left(\mathrm{Law}((R_{b,u+t})_{t\geq 0}), \mathrm{Law}((S_{u+t})_{t\geq 0})\right) \\ &\qquad \leq 2\varepsilon + \sup_{M,N\in\mathcal{K}_\varepsilon} d_{\mathrm{TV}}(P_u^M, P_u^N). \end{aligned} \tag{6.37}$$

The last term tends to zero as $u \to \infty$, since both $M \mapsto M(0)$ and the norm are bounded on a compact subset of $\mathcal{E}_{\lambda,m}$. Letting $\varepsilon \downarrow 0$ proves (6.33). □

Fix $T \in \mathbb{R}$ and let

$$\mu_{b,+}^T \overset{\text{def}}{=} \mathrm{Law}\big((S_{\sigma_b^+ + t} - S_{\sigma_b^+})_{t\geq T}\big). \tag{6.38}$$

Proposition 6.7 (Cauchy property in total variation)**.** *For each $T \in \mathbb{R}$,*

$$\lim_{B\to\infty} \sup_{b,c\geq B} d_{\mathrm{TV}}(\mu_{b,+}^T, \mu_{c,+}^T) = 0. \tag{6.39}$$

Proof. Fix $\varepsilon > 0$. By Proposition 6.6, choose $u > 0$ and then $b_0 < \infty$ such that, for $b \geq b_0$, there is a coupling of R_b and S for which

$$R_{b,u+t} = S_{u+t} \quad \text{for every } t \geq 0 \tag{6.40}$$

with probability at least $1 - \varepsilon$. Such a coupling of the full paths follows from a maximal coupling of their tail projections and regular conditional laws; see Lemma B.2.

Put

$$L \stackrel{\text{def}}{=} u + \max\{0, -T\}. \tag{6.41}$$

For $c > 0$, define the second positive hitting time

$$\rho_{b,c} \stackrel{\text{def}}{=} \inf\{t \geq 0 : R_{b,t} - at = -c\}. \tag{6.42}$$

By compact-path tightness in Proposition 6.5,

$$\lim_{c\to\infty} \limsup_{b\to\infty} \mathbb{P}[\rho_{b,c} < L] = 0. \tag{6.43}$$

Indeed, the event $\{\rho_{b,c} < L\}$ implies

$$\inf_{0\leq t\leq L} (R_{b,t} - at) \leq -c,$$

and the displayed infima form a tight family. Similarly,

$$\mathbb{P}[\sigma_c^+ < L] \longrightarrow 0 \qquad (c \to \infty). \tag{6.44}$$

Choose c_0 so that the probabilities in (6.43) and (6.44) are at most ε for $b \geq b_0$ and $c \geq c_0$. On the event where (6.40) holds and both hitting times are at least L, the two drifted paths agree from time u onward and neither reaches level $-c$ before time L. Therefore

$$\rho_{b,c} = \sigma_c^+. \tag{6.45}$$

Moreover, if $t \geq T$, then $\rho_{b,c} + t \geq u$, so

$$R_{b,\rho_{b,c}+t} - R_{b,\rho_{b,c}} = S_{\sigma_c^+ + t} - S_{\sigma_c^+}. \tag{6.46}$$

Positive first-passage additivity gives

$$\rho_b + \rho_{b,c} = \rho_{b+c}; \tag{6.47}$$

see Lemma B.1. Thus the left side of (6.46) has law $\mu^T_{b+c,+}$, whereas the right side has law $\mu^T_{c,+}$. We obtain

$$\lim_{B\to\infty} \sup_{b,c\geq B} d_{\mathrm{TV}}(\mu^T_{b+c,+}, \mu^T_{c,+}) = 0. \tag{6.48}$$

For $b, c \geq B$,

$$d_{\mathrm{TV}}(\mu^T_{b,+}, \mu^T_{c,+}) \leq d_{\mathrm{TV}}(\mu^T_{b,+}, \mu^T_{b+c,+}) + d_{\mathrm{TV}}(\mu^T_{b+c,+}, \mu^T_{c,+}).$$

Apply (6.48) to the ordered pairs (c, b) and (b, c) to prove (6.39). □

Proof of Theorem B. For fixed T, Proposition 6.7 shows that $(\mu^T_{b,+})_{b>0}$ is Cauchy in total variation. Finite signed measures are complete in the total variation norm, so there is a probability measure μ^T_∞ on $C([T, \infty))$ such that

$$d_{\mathrm{TV}}(\mu^T_{b,+}, \mu^T_\infty) \longrightarrow 0. \tag{6.49}$$

If $T_1 < T_2$, the restriction of $\mu^{T_1}_\infty$ to $[T_2, \infty)$ is $\mu^{T_2}_\infty$, since the same is true before taking the limit and total variation distance decreases under restriction. The compatible laws $(\mu^T_\infty)_{T\in\mathbb{R}}$ therefore determine a probability law on $C(\mathbb{R})$; see Lemma B.3. Let S_∞ have this law.

By (6.25), for every $T \in \mathbb{R}$,

$$\begin{aligned} d_{\mathrm{TV}}\Big(\mathrm{Law}((S_{\sigma_b+t} - S_{\sigma_b})_{t\geq T}), \mathrm{Law}((S_{\sigma_b^+ + t} - S_{\sigma_b^+})_{t\geq T})\Big) \\ \leq \mathbb{P}[\sigma_b \neq \sigma_b^+] \longrightarrow 0. \end{aligned} \tag{6.50}$$

Combining (6.49) and (6.50) proves the total variation convergence in Theorem B. Since every recentered path vanishes at zero, $S_{\infty,0} = 0$ almost surely. □

6.3 The field local limit

We now combine Theorem B with the radial–spherical decomposition. We first verify that the radial process in Definition 2.1 gives a random distribution. For $T \in \mathbb{R}$, let $\mathcal{R}_T$ be the map which sends a continuous function f on $[T, \infty)$ to the radially symmetric distribution on $B(0, \mathrm{e}^{-T})$ whose value on $\partial B(0, \mathrm{e}^{-s})$ is $f(s) + \beta s$. For $\varphi \in C_c^\infty(B(0, \mathrm{e}^{-T}))$,

$$\langle \mathcal{R}_T(f), \varphi \rangle = |\mathbb{S}^{d-1}| \int_T^\infty (f(s) + \beta s) \left(\int_{\mathbb{S}^{d-1}} \varphi(\mathrm{e}^{-s}\theta) \, \mathrm{d}\sigma_{d-1}(\theta) \right) \mathrm{e}^{-ds} \, \mathrm{d}s. \tag{6.51}$$

On the Borel set of functions of subexponential growth, all pairings in (6.51) converge and define a distribution; extend the map arbitrarily outside this set. Lemma B.4 shows that the resulting map is Borel measurable. Each recentred path is sublinear and therefore belongs to the Borel set $\mathcal{X}_T$ from Lemma B.4. Total variation convergence implies that the limiting law also gives $\mathcal{X}_T$ full mass. Hence the arbitrary extension is not used.

Proof of Theorem C. By translation invariance, it is enough to consider $z = 0$. Write

$$h = h_{\mathrm{rad}} + h_{\mathrm{sph}} \tag{6.52}$$

for the decomposition in Lemma 3.1. The time σ_b is measurable with respect to h_{rad}. Since dilations preserve $\mathcal{H}_{d,\mathrm{sph}}$, conditioning on σ_b and using independence in (6.52) gives

$$h_{\mathrm{sph}}(\mathrm{e}^{-\sigma_b}\cdot) \overset{\mathrm{law}}{=} h_{\mathrm{sph}}, \tag{6.53}$$

and the field on the left remains independent of the recentered radial process.

The spherical average of $h^{\sigma_b, b}$ on the sphere of radius e^{-s} is

$$\begin{aligned} h^{\sigma_b, b}_{\mathrm{e}^{-s}}(0) &= S_{\sigma_b + s} + \beta(\sigma_b + s) - Q\sigma_b + b \\ &= S_{\sigma_b + s} - S_{\sigma_b} + \beta s, \end{aligned} \tag{6.54}$$

where we used $S_{\sigma_b} - a\sigma_b = -b$ and $a = Q - \beta$. Fix $R > 0$ and set $T = -\log R$. By (6.54), the radial restriction of $h^{\sigma_b, b}$ to $B(0, R)$ is the image under $\mathcal{R}_T$ of $(S_{\sigma_b + s} - S_{\sigma_b})_{s \geq T}$. Theorem B therefore gives convergence of these radial restrictions in total variation.

Taking the product with the fixed independent law in (6.53) does not change the total variation distance. Applying the measurable addition map from the pair of radial and spherical distributions to their sum can only decrease total variation distance. The limiting field is exactly the field $h^\star$ in Definition 2.1. This proves Theorem C. □

A Analytic facts

We record the standard facts concerning Cameron–Martin shifts and relative entropy which were used in Section 5. They are stated on the infinite time interval because the comparison in Proposition 5.9 concerns the entire future path.

Lemma A.1 (Cameron–Martin shift on the half-line)**.** *Let W be Wiener measure on $C([0, \infty))$, and let $g \in L^2(0, \infty)$. Put*

$$h_g(t) \overset{\mathrm{def}}{=} \int_0^t g(s) \, \mathrm{d}s. \tag{A.1}$$

If W_g is the image of W under $\omega \mapsto \omega + h_g$, then W_g is absolutely continuous with respect to W and

$$\operatorname{Ent}(W_g \mid W) = \frac{1}{2}\|g\|^2_{L^2(0,\infty)}. \tag{A.2}$$

If Φ is a measurable map into a standard Borel space, then

$$\operatorname{Ent}(\Phi_* W_g \mid \Phi_* W) \le \operatorname{Ent}(W_g \mid W) \tag{A.3}$$

and

$$d_{\mathrm{TV}}(\Phi_* W_g, \Phi_* W) \le \sqrt{\frac{1}{2}\operatorname{Ent}(\Phi_* W_g \mid \Phi_* W)} \le \frac{1}{2}\|g\|_{L^2(0,\infty)}. \tag{A.4}$$

Proof. Apply the Cameron–Martin formula on $[0, T]$. The Radon–Nikodym derivative is

$$\exp\left(\int_0^T g(s)\,\mathrm{d}W_s - \frac{1}{2}\int_0^T g(s)^2\,\mathrm{d}s\right), \tag{A.5}$$

and the entropy is $\frac{1}{2}\int_0^T g(s)^2\,\mathrm{d}s$. Since $g \in L^2(0,\infty)$, the exponential martingales in (A.5) are uniformly integrable after the standard localisation argument, and the finite-horizon Radon–Nikodym derivatives converge to the half-line Cameron–Martin density. Monotone convergence of the energies gives (A.2). The data-processing inequality gives (A.3), and Pinsker's inequality gives (A.4); see, for example, [3, Chapter 2]. □

Lemma A.2 (Weighted point evaluation)**.** *For $M \in \mathcal{E}_{\lambda,m}$, $u \ge 2$, and $0 \le k \le m$,*

$$|M^{(k)}(u)| \le C_{m,\lambda} \mathrm{e}^{-\lambda u}\|M\|_{\mathcal{E}_{\lambda,m}}. \tag{A.6}$$

Proof. Set $F_k(r) = \mathrm{e}^{\lambda r} M^{(k)}(r)$. On $[u-1, u+1]$, the one-dimensional Sobolev estimate gives

$$|F_k(u)| \le C\big(\|F_k\|_{L^2(u-1,u+1)} + \|F_k'\|_{L^2(u-1,u+1)}\big).$$

Since $F_k' = \mathrm{e}^{\lambda r}(\lambda M^{(k)} + M^{(k+1)})$, the right side is at most $C_{m,\lambda}\|M\|_{\mathcal{E}_{\lambda,m}}$. Multiplication by $\mathrm{e}^{-\lambda u}$ gives (A.6). □

B Path-space facts

Lemma B.1 (First-passage additivity)**.** *Let $x \in C([0,\infty))$ satisfy $x(0) = 0$ and $x(t) \to -\infty$ as $t \to \infty$. For $b > 0$, put*

$$T_b \stackrel{\mathrm{def}}{=} \inf\{t \ge 0 : x(t) = -b\}. \tag{B.1}$$

For $c > 0$, let

$$T_{b,c} \stackrel{\mathrm{def}}{=} \inf\{t \ge 0 : x(T_b + t) - x(T_b) = -c\}. \tag{B.2}$$

Then

$$T_b + T_{b,c} = T_{b+c}. \tag{B.3}$$

Proof. For $0 \le t < T_{b,c}$, one has $x(T_b + t) - x(T_b) > -c$, hence $x(T_b + t) > -(b + c)$. Also, by the definition of T_b, $x(t) > -b > -(b + c)$ for $0 \le t < T_b$. At time $T_b + T_{b,c}$, continuity gives $x(T_b + T_{b,c}) = -(b + c)$. This is exactly (B.3). □

Lemma B.2 (Coupling full paths from a tail coupling)**.** *Let $E = C([T_0, \infty))$, let $T_1 > T_0$, and let $r : E \to C([T_1, \infty))$ be restriction. Suppose that μ and ν are probability measures on E. There is a coupling (X, Y) of (μ, ν) such that*

$$\mathbb{P}[r(X) = r(Y)] = 1 - d_{\mathrm{TV}}(r_*\mu, r_*\nu). \tag{B.4}$$

Proof. The path spaces are Polish. First take a maximal coupling (U, V) of the two tail laws $r_*\mu$ and $r_*\nu$. Choose regular conditional probabilities $\mu(\,\mathrm{d}x \mid r(x) = U)$ and $\nu(\,\mathrm{d}y \mid r(y) = V)$, and sample X and Y conditionally and independently from these kernels. Their marginal laws are μ and ν, and $r(X) = U$, $r(Y) = V$, which gives (B.4). □

Lemma B.3 (Compatible tail laws)**.** *For each integer $n \geq 1$, let μ_n be a probability measure on $C([-n, \infty))$. Suppose that, whenever $n < k$, the restriction of μ_k to $[-n, \infty)$ is μ_n. Then there is a unique probability measure μ on $C(\mathbb{R})$ whose restriction to $[-n, \infty)$ is μ_n for every n.*

Proof. The restriction maps make $(C([-n, \infty)))_{n \geq 1}$ a projective system of Polish spaces. Its projective limit is naturally identified with $C(\mathbb{R})$: a compatible sequence of continuous tail paths pastes to a unique continuous path on $\mathbb{R}$. The usual extension theorem for consistent probability measures on standard Borel spaces therefore gives a probability measure on this projective limit. Its finite tail restrictions determine it uniquely. □

Lemma B.4 (Measurability of the radial reconstruction)**.** *Fix $T \in \mathbb{R}$. Let $\mathcal{X}_T \subset C([T, \infty))$ be the set of functions f such that*

$$\limsup_{s \to \infty} \frac{\log(1 + |f(s)|)}{s} < d. \tag{B.5}$$

Then $\mathcal{X}_T$ is Borel, and the map $\mathcal{R}_T$ defined by (6.51) is Borel measurable from $\mathcal{X}_T$ to $\mathcal{D}'(B(0, \mathrm{e}^{-T}))$. Extending $\mathcal{R}_T$ by zero outside $\mathcal{X}_T$ gives a Borel map on all of $C([T, \infty))$.

Proof. Condition (B.5) can be written using countably many bounds on the intervals $[n, n+1]$, so $\mathcal{X}_T$ is Borel. For fixed $\varphi \in C_c^\infty(B(0, \mathrm{e}^{-T}))$, the integral in (6.51) is the limit, as $N \to \infty$, of the integral over $[T, N]$. The truncated pairing is continuous in the topology of uniform convergence on compact sets. Hence each pairing is Borel measurable on $\mathcal{X}_T$.

Choose a countable dense set of test functions in $C_c^\infty(B(0, \mathrm{e}^{-T}))$ for its usual nuclear Fréchet topology. The Borel σ-field on $\mathcal{D}'(B(0, \mathrm{e}^{-T}))$ is generated by evaluation on this set. Therefore the distribution-valued map is Borel measurable. Sublinear functions satisfy (B.5), so this applies to all paths used in Section 6. □

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